

Advanced Machine Learning
Statistical Learning Theory
Generalization Bounds
Uniform stability

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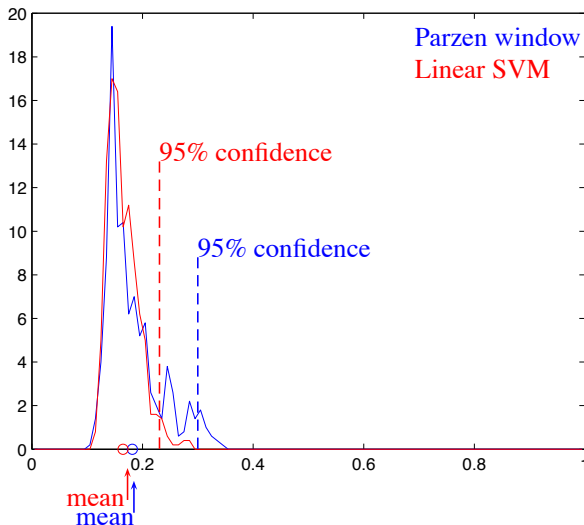
Statistical Learning Theory - Context

Context

For a fixed algorithm, function class and sample size, one considers generating random samples $\rightarrow$ leads to a distribution of errors

- Focusing on the mean of the error distribution?
 $\rightarrow$ can be misleading: the learner has only **one** sample
- **Statistical Learning Theory**: **tail of distributions**
 $\rightarrow$ finding bounds which hold with high probability over random samples of size m
- Compare to a statistical test at 99% confidence level $\rightarrow$ chances of the conclusion not being true is less than 1%
- We rather consider the **PAC** framework
Use of a **confidence parameter** δ : $P^m[\text{large error}] \leq \delta$
 δ is probability of being misled by the training set
- Hence **high confidence**: $P^m[\text{approximately correct}] \geq 1 - \delta$

Error distribution picture



Distribution of test errors with respect to different (random) learning samples

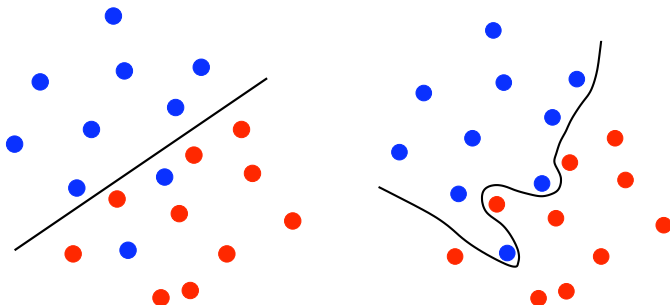
Credit: J. Shawe-Taylor, NeurIPS'18 Tutorial: Statistical Learning Theory: A Hitchhiker's Guide

In this course

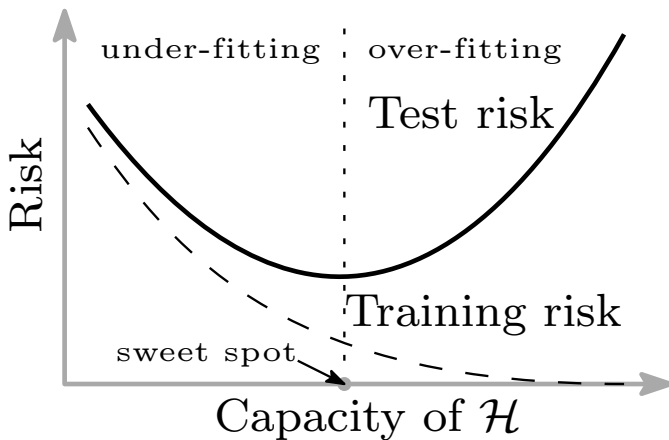
- We give an overview of main goals, methods and ideas
- We will outline some proof principles
- A kind of Hitchhiker's guide! ;-)
- But we will not: provide all detailed proofs/full literature and give a full coverage of full literature and paradigms
- We will derive completely the foundations of one framework and detailed one particular proof
- Resources
 - Tutorial of J. Shawe-Taylor and O. Rivasplata at NeurIPS'18
 - Paper on Uniform stability of O. Bousquet and E. Eliseef, JMLR 2002
 - Any statistical learning theory course
 - Other papers for some specific parts of this course
 - Paper on *Robustness and Generalization* of H. Xu and S. Mannor, Machine Learning, 2012. <https://arxiv.org/abs/1005.2243>
 - Survey on PAC-Bayes theory from B. Guedj et al. *A Primer on PAC-Bayesian Learning*, 2019. <https://arxiv.org/pdf/1901.05353.pdf>

Learning is to be able to generalize

- How to characterize learning from examples?
- Memorising the data already seen is a bad approach: **overfitting**
- **Generalization** is the ability to perform well on **unseen data**



A famous illustration of Generalization



from: M. Belkin et al. Reconciling modern machine-learning practice and the classical bias–variance trade-off. Proceedings of the National Academy of Sciences, 116(32):15849–15854, 2019.

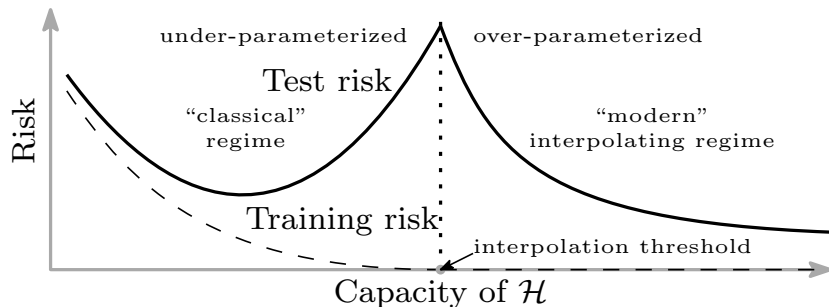
What about deep learning?

Deep Neural networks architectures trained on extremely large datasets achieve almost zero training error ...

while achieving very strong performances on tests sets: this is not overfitting.

... But requiring millions of annotated data, GPU extensive training, which is expensive with significant environmental footprint.

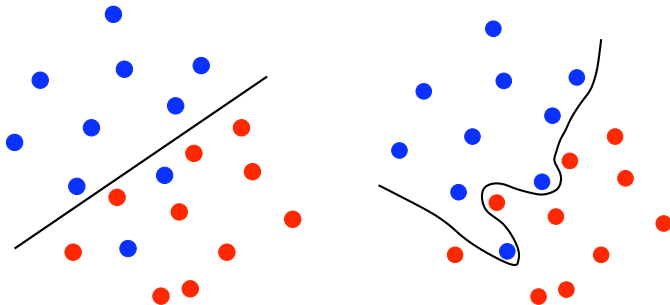
we may only get half of the picture



from: M. Belkin et al. Reconciling modern machine-learning practice and the classical bias–variance trade-off. Proceedings of the National Academy of Sciences, 116(32):15849–15854, 2019.

Learning is to be able to generalize

- But not from scratch! Each learning task must not be considered as a new fresh one but use existing context.
- Need to include bias: structure/semantic information/implicit representations of the world
- Design better algorithms (more "intelligent", frugal/resource-efficient, etc.)



Supervised classification: notations

- $\mathcal{X}$: input space, Y label space $\mathcal{Z} : \mathcal{X} \times Y$
- We are given a **training set** S of m labeled instances

$$S = \{\mathbf{z}_i = (\mathbf{x}_i, y_i)\}_{i=1}^m \subset (\mathcal{X} \times Y)^m$$

where $\mathbf{z}_i$'s follow some (unknown) distribution P , $\mathbf{x}_i$'s lie in some **feature space** $\mathcal{X}$ (e.g., $\mathbb{R}^d$) and y_i 's are the output, $y_i \in Y$.

- We focus here on **binary classification**: $\forall i, y_i \in \{-1, 1\} = Y$.
- We are given a class of hypothesis $\mathcal{H} \subseteq Y^{\mathcal{X}}$
- We have a loss function $\ell : \mathcal{H} \times \mathcal{Z} \rightarrow \mathbb{R}^+$, $\mathcal{Z} = \mathcal{X} \times Y$, ℓ is assumed to be bounded by M
- True risk (generalization error - out-of-sample): $\mathcal{R}(h) = \mathbb{E}_{\mathbf{z} \sim P} \ell(h, \mathbf{z})$
- Empirical risk (in-sample) $\hat{\mathcal{R}}(h) = \hat{\mathcal{R}}_S(h) = \frac{1}{m} \sum_{i=1}^m \ell(h, \mathbf{z}_i)$

Statistical Learning Assumptions

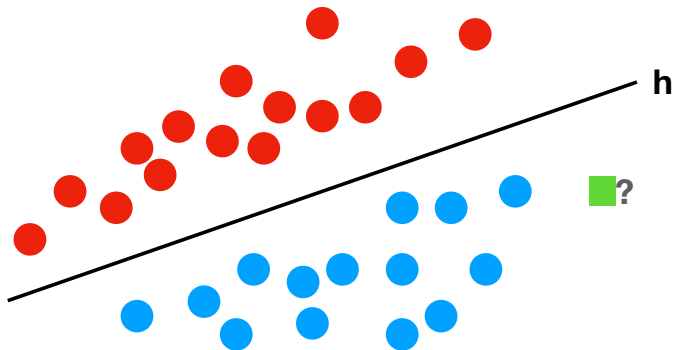
- P is a data generating distribution over $\mathcal{Z}$
- A learning algorithm $A : \mathcal{Z}^m \rightarrow \mathcal{H}$
- The learner does not know P , it can only see a training sample.
- The training instances are *identically and independently distributed (i.i.d.)* from P .

Some elements above can be relaxed (but beyond the scope of this course)

Supervised classification: goal

Goal

Using S , learn a classifier $h : \mathcal{X} \rightarrow \{-1, 1\}$ that, given a new instance drawn from P , is able to predict its label (with low error).



What to achieve from the sample

We use the available sample to

- 1 Learn a predictor
- 2 Certify the predictor's performance

Learning a predictor

- algorithm driven by some learning principle
- informed by prior knowledge resulting in inductive bias

Certifying performance

- What happens beyond the training set
- Generalization bounds

Actually the two goals interact with each other!

Generalization

If a classifier h does well on the in-sample S ...will it still do well on out-of-sample instances?

- Generalization gap: $\Delta(h) = \mathcal{R}(h) - \hat{\mathcal{R}}(h)$
- Upper bounds: with high probability (w.h.p.) $\Delta(h) \leq \epsilon(m, \delta)$
 $\Rightarrow \mathcal{R}(h) \leq \hat{\mathcal{R}}(h) + \epsilon(m, \delta)$
- Lower bounds: w.h.p. $\Delta(h) \geq \tilde{\epsilon}(m, \delta)$

Settings:

- distribution-free $\neq$ distribution-dependent
- algorithm-free $\neq$ algorithm-dependent

PAC Learning

Probably Approximately Correct - PAC [Valiant 84]

- Let h be an hypothesis learned by an algorithm A on a learning sample of size m
- $h^* = \operatorname{argmin}_{h \in \mathcal{H}} R(h)$ best hypothesis in class
- For any $\epsilon > 0$, any δ , for all distributions P on $\mathcal{X} \times \mathcal{Y}$ with $\mathcal{X} \subseteq \mathbb{R}^d$, the following holds for any sample size $\geq \operatorname{poly}(1/\epsilon, 1/\delta, d)$:

$$\mathcal{P} \left(\left| \hat{R}(h) - R(h^*) \right| \geq \epsilon \right) \leq \delta$$

- A tries generally to minimize the empirical risk often w.r.t. some regularization
- If the probability is small, we can have good approximations of the target function (if $R(h^*) = 0$ the concept is PAC-learnable)
- if $R(h^*)$ is (very) small, the concept can be learned by h , we would like something even stronger (Uniform convergence):

$$\mathcal{P} \left(\sup_{h \in \mathcal{H}} \left| \hat{R}(h) - R(h^*) \right| \geq \epsilon \right) \leq \delta$$

The (weak) law of large numbers

If $x_1, x_2, \dots, x_m$ are m *independently and identically distributed (i.i.d.)* samples of a random variable X distributed by P . Then for a small positive non-zero value ϵ :

$$\lim_{m \rightarrow \infty} \mathcal{P} \left[\left| \mathbb{E}_{X \sim P}[X] - \frac{1}{m} \sum_{i=1}^m x_i \right| > \epsilon \right] = 0$$

- By noting that $\hat{R}$ is the sample mean of the errors and the true risk is the true mean, we can apply the Thm on a single hypothesis:
$$\lim_{m \rightarrow \infty} \mathcal{P} \left(\left| \hat{R}(h) - R(h) \right| \geq \epsilon \right) = 0$$
- Note the *i.i.d.* restriction
- No information about the convergence (nb of examples needed to achieve the result)

Concentration Inequality: Hoeffding bound

Concentration inequalities: quantify how much random variables deviate from their expected values.

Hoeffding bound

Let $Z_1, \dots, Z_m$ be a sequence of *i.i.d* random variables and let $\bar{Z} = \frac{1}{m} \sum_{j=1}^m Z_j$. Assume that $\mathbb{E}[\bar{Z}] = \mu$ and for all $1 \leq i \leq m$ we have $a \leq Z_j \leq b$. Then, for any $\epsilon > 0$:

$$P \left[\left| \frac{1}{m} \sum_{j=1}^m Z_j - \mu \right| > \epsilon \right] \leq 2 \exp \left(-2m\epsilon^2 / (b - a)^2 \right).$$

- Assuming the errors (the loss) $\in [0, 1]$, for a single hypothesis we can get:

$$\mathcal{P} \left(\left| \hat{R}(h) - R(h) \right| \geq \epsilon \right) \leq 2 \exp(-2m\epsilon^2)$$

Building block: One single function

- We consider one fixed (non-data dependent) h

$$\mathbb{E}_{S \sim P^m}[\hat{\mathcal{R}}_S(h)] = \mathcal{R}(h)$$

- $P^m[\Delta(h) > \epsilon] = P^m[\mathbb{E}_{S' \sim P^m}[\hat{\mathcal{R}}_{S'}(h)] - \hat{\mathcal{R}}_S(h) > \epsilon]$
- $\ell(h(x_i), y_i)$ are independent random variable
- If $0 \leq \ell(h(x_i), y_i) \leq 1$ (can be generalized to any B -bounded loss), with **Hoeffding inequality**:

$$P^m[\Delta(h) > \epsilon] \leq \exp(-2m\epsilon^2) = \delta$$

- Given $\delta \in (0, 1)$, equate RHS to δ , solve equation for ϵ , get:

$$P^m[\Delta(h) > \sqrt{(1/2m) \log(1/\delta)}]$$

- Then, with probability $\geq 1 - \delta$

$$\mathcal{R}(h) \leq \hat{\mathcal{R}}_S(h) + \sqrt{\frac{\log(1/\delta)}{2m}}$$

When $\mathcal{H}$ is finite

Union Bound

- Let $A_1, A_2, \dots, A_k$ be k events (not necessarily independent) then:
$$P(A_1 \cup A_2 \cup \dots \cup A_k) \leq P(A_1) + P(A_2) + \dots + P(A_k)$$
- The PAC learning equation can be reformulated as follows:

$$\mathcal{P} \left(\sup_{h \in \mathcal{H}} |\hat{R}(h) - R(h)| \geq \epsilon \right) = \mathcal{P} \left(\bigcup_{h \in \mathcal{H}} |\hat{R}(h) - R(h)| \geq \epsilon \right)$$

- Thus, from union bound

$$\mathcal{P} \left(\bigcup_{h \in \mathcal{H}} |\hat{R}(h) - R(h)| \geq \epsilon \right) \leq \sum_{h \in \mathcal{H}} \mathcal{P} \left(|\hat{R}(h) - R(h)| \geq \epsilon \right)$$

Bound when $\mathcal{H}$ finite: $|\mathcal{H}| < \infty$

- We would like a uniform bound: $P^m[\forall h \in \mathcal{H}, \Delta(h) \leq \epsilon] \geq 1 - \delta$
- Using the fact that for one hypothesis we have

$$\mathcal{P}\left(\left|\hat{\mathcal{R}}(h) - \mathcal{R}(h)\right| \geq \epsilon\right) \leq 2 \exp(-2m\epsilon^2)$$

we can get:

$$\mathcal{P}\left(\sup_{h \in \mathcal{H}} \left|\hat{\mathcal{R}}(h) - \mathcal{R}(h)\right| \geq \epsilon\right) \leq 2|\mathcal{H}| \exp(-2m\epsilon^2)$$

- Fixing $\delta = 2|\mathcal{H}| \exp(-2m\epsilon^2)$, we get that $\epsilon = \sqrt{\frac{\ln(|\mathcal{H}|) + \ln \frac{2}{\delta}}{2m}}$
- We can then state that with probability $1 - \delta$:

$$\mathcal{R}(h) \leq \hat{\mathcal{R}}(h) + \sqrt{\frac{\ln(|\mathcal{H}|) + \ln\left(\frac{2}{\delta}\right)}{2m}}$$

- Not sufficient when $\mathcal{H}$ is infinite.

Bias-Variance result

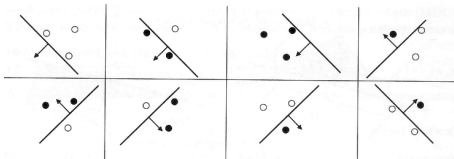
When $h = \operatorname{argmin}_h \hat{\mathcal{R}}(h)$, we can get a bias-variance based result

$$\begin{aligned}\mathcal{R}(h) &\leq \hat{\mathcal{R}}(h) + \epsilon \\ &\leq \hat{\mathcal{R}}(h^*) + \epsilon \text{ because } h = \operatorname{argmin}_h \hat{\mathcal{R}}(h) \\ &< \mathcal{R}(h^*) + \epsilon + \epsilon \text{ applying inequality on the other way around} \\ &= \mathcal{R}(h^*) + 2\sqrt{\frac{\ln(|\mathcal{H}|) + \ln(\frac{2}{\delta})}{2m}}\end{aligned}$$

- $\mathcal{R}(h^*)$ plays the role of the bias, second right hand side term is related to variance
- Does not work when $\mathcal{H}$ is infinite.

Uncountably infinite function class: $|\mathcal{H}| \geq \mathbb{N}$

- **Double sample trick:** a second "ghost" sample
 - True error $\leftrightarrow$ empirical error
 - hence reduce to a finite number of behaviours
 - make union bound, but bad events grouped together
- **Symmetrization**
 - bound the probability of good performance on one sample but bad performance on the other sample
 - swapping examples between actual and ghost sample
- **Growth function** of class $\mathcal{H}$:
 - $G_{\mathcal{H}}(m)$: largest number of dichotomies (± 1 labels) generated by the class $\mathcal{H}$ on any m points
- **VC dimension** of class $\mathcal{H}$
 $VC(\mathcal{H})$: largest m such that $G_{\mathcal{H}}(m) = 2^m$



Double sample trick

We fix an error ϵ . Let S be a sample of $m > 4 \ln 2 / \epsilon^2$ i.i.d. from D . Let S' be a second sample of m points, i.i.d. from D , called the ghost sample (not provided to a learning algorithm).

Intuition

Taking the error of a hypothesis on the ghost sample as a proxy for the generalization error of the hypothesis. This allows us to make claims about a hypothesis being ϵ -bad without examining every hypothesis in an infinite hypothesis space. With the double sample trick, we can focus on what is happening on finite sets of points without worrying about what is happening over the entire domain.

Double sample trick

- Let $M(h, S)$: nb of mistakes h makes on S .
- Let $B : [\exists h \in \mathcal{H} : h \text{ consistent with } S \text{ and } h \text{ } \epsilon\text{-bad on } D]$, we want to prove that $Pr[B] \leq \delta$
- Let $B' : [\exists h \in \mathcal{H} : h \text{ consistent with } S \text{ and } M(h, S') > \frac{m\epsilon}{2}]$

Step 1: $Pr[B'|B] \geq 1/2$

The same hypothesis h that makes B true makes B' true with probability at least $1/2$. Suppose h is consistent with S and has $err(h) > \epsilon$. Then we must show with probability at least $1/2$ that $M(h, S') > \frac{m\epsilon}{2}$.

$\mathbb{E}[M(h, S')] > m\epsilon$ since $err(h) > \epsilon$. Therefore by Chernoff bounds, we have that $Pr[M(h, S') \leq \frac{m\epsilon}{2}] \leq 1/2$.

Chernoff: $Pr(\hat{X} < (1 - \delta)\mu) \leq e^{-\delta^2\mu^2/m}$, take $\delta = 1/2$ and $\mu = m\epsilon$, use $m \geq 4 \ln 2 / \epsilon^2$

Put in another way, if B is true, then the probability that B' is not true is less than $1/2$. Thus, $Pr[B'|B] \geq 1/2$.

Step 2: $Pr[B] \leq 2Pr[B']$

Trivially, we have $B \cap B' \Rightarrow B'$

Then, $Pr[B'] \geq Pr[B \cap B'] = Pr[B]Pr[B'|B] \geq \frac{1}{2}Pr[B]$ by step 1.

So $Pr[B] \leq 2Pr[B']$.

Therefore, it is sufficient to find an upper bound on $Pr[B']$

break

At this point we look at some intuition for why B' has low probability. B' entails having no error on S and lots of error on S' . Since the samples were chosen independently, all permutations of the data among the sets S and S' are equally likely, so it is unlikely that all errors are in S' .

Consider two experiments for generating S and S' . Experiment 1: Choose S and S' as usual, independently from D .

Experiment 2: First choose S and S' as usual, then move examples around as follows.

For each $i \in \{1, \dots, m\}$, with 50% probability swap element i of S with element i of S' . Call the resulting samples T and T' .

Note that the distributions for T , T' exactly equals those for S , S' . Now define

$$B'' : [\exists h \in \mathcal{H} : h \text{ consistent with } T \text{ and } M(h, T') > \frac{m\epsilon}{2}]$$

Step 3: $Pr[B''] = Pr[B']$

Since the distributions for T, T' exactly equals those for S, S' we conclude $Pr[B''] = Pr[B']$

Define $b(h) : [h \text{ consistent with } T \text{ and } M(h, T') > \frac{m\epsilon}{2}]$

Step 4: $Pr[b(h)|S, S'] \leq 2^{-m\epsilon/2}$

In step 4, we select S and S' according to the standard procedure and then look at $b(h)$ before proceeding to construct T, T' according to experiment 2.

We show through three cases that $Pr[b(h)|S, S'] \leq 2^{-m\epsilon/2}$.

Case 1: h produces errors on two corresponding samples in S and S' .

In this case, regardless of the permutations of the data selected by experiment 2, $b(h)$ is false. So $Pr[b(h)|S, S'] = 0$.

To illustrate Case 1, we represent in the following table example sets S and S' . A zero in the i th column indicates that the label of the i th element is correct, and a one in the i th column indicates that the label of the i th element is incorrect. Experiment 2 allows for an element to be swapped with the element below it. In this example there will always be an error in the second row because of the column of ones.

S	1	0	1
S'	1	1	0

Let r denote the number of points in S with label different from the corresponding point in S' .

Case 2: $r < \frac{m\epsilon}{2}$

In this case, $b(h)$ cannot happen because there are not enough mistakes. In order for $b(h)$ to be true, all mistakes must occur in S' and none may occur in S and the total number of errors must exceed $\frac{m\epsilon}{2}$. Since there are insufficient errors for this to occur, we again have $Pr[b(h)|S, S'] = 0$

Case 3: $r \geq \frac{m\epsilon}{2}$

In this case, for $b(h)$ to be true, all r errors must be placed in T' rather than T . These events happen independently, each with probability $1/2$. So $Pr[b(h)|S, S'] = 2^{-r} \leq 2^{-m\epsilon/2}$.

In all three cases $Pr[b(h)|S, S'] \leq 2^{-m\epsilon/2}$.

Step 5: $Pr[B''] \leq G_{\mathcal{H}}(2m)2^{-m\epsilon/2}$

We now go to bound the probability of B'' . The number of behaviors a hypothesis can take on over the $2m$ samples in T , T' is finite, given by $G_{\mathcal{H}}(2m)$. For each behavior, we select a single representative hypothesis $h \in \mathcal{H}$ giving that behavior, giving a set $\mathcal{H}'(S, S')$ among $G_{\mathcal{H}}(2m)$ representative hypotheses.

To reach the second line that follows we will use marginalization ($Pr[A] = \mathbb{E}_X[Pr[A|X]]$). We can go from the second to the third line since each hypothesis $h \in \mathcal{H}$ has the same behavior on S, S' as some other $h \in \mathcal{H}'(S, S')$.

$$Pr[B''] = Pr[\exists h \in \mathcal{H} : b(h)] \quad (1)$$

$$= \mathbb{E}_{S, S'} Pr[\exists h \in \mathcal{H} : b(h) | S, S'] \text{ using marginalization} \quad (2)$$

$$= \mathbb{E}_{S, S'} Pr[\exists h \in \mathcal{H}'(S, S') : b(h) | S, S'] \text{ as explained above} \quad (3)$$

$$\leq \mathbb{E}_{S, S'} \sum_{h \in \mathcal{H}'(S, S')} Pr[b(h) | S, S'] \text{ by union bound} \quad (4)$$

$$\leq \mathbb{E}_{S, S'} G_{\mathcal{H}}(2m) 2^{-m\epsilon/2} \quad (5)$$

Lastly, we finally get a bound on $Pr[B]$:

$$Pr[B] \leq 2Pr[B'] = 2Pr[B''] \leq 2G_{\mathcal{H}}(2m) 2^{-m\epsilon/2} \leq \delta.$$

IF some explicitly for ϵ , we have $\epsilon \leq 2 \frac{\ln G_{\mathcal{H}}(2m) + \ln(2/\delta) + 1}{m}$

Examples - What is the VC-dimension?

Example 1

- $\mathcal{X} = \mathbb{R}$, the real line.
- $\mathcal{H}$ is the set of positive half-lines.

Clearly $\{x\}$ can be shattered (by choosing half-lines starting before and after x we get different behaviors). Also if $x < y$ are two real numbers, then we can never get the behavior $(h(x) = +1, h(y) = -1)$. Thus no set of size > 1 can be shattered and so the VC-dimension is 1.

Examples - What is the VC-dimension?

Example 2

- $\mathcal{X} = \mathbb{R}$, the real line.
- $\mathcal{H}$ is the set of intervals.

Here $\{x, y\}$ can clearly be shattered. But if $x < y < z$ are three real numbers then $h(x) = h(z) = +1$ and $h(y) = -1$ can never occur, thus a set of size > 2 cannot be shattered. Thus the VC-dimension is 2.

Examples - What is the VC-dimension?

Example 3

- $\mathcal{X} = \mathbb{R}^2$.
- $\mathcal{H}$ is the set of axis parallel rectangles.

Showing the VC dimension is at least 4 is easy – consider 4 points in the shape of a diamond (say we look at $(1, 0)$, $(-1, 0)$, $(0, 1)$, $(0, -1)$). It is clear this set can be shattered. Now suppose we have a set of 5 points. Consider the bottom-most, top-most, left-most and right-most points (they may coincide). We cannot have the behavior that h is $+1$ on these points and -1 on the remaining. Thus no set of size ≥ 5 can be shattered, the VC-dimension is then 4.

Examples - What is the VC-dimension?

Example 4

- $\mathcal{X} = \mathbb{R}^d$.
- $\mathcal{H}$ is the set of half-spaces.

The VC-dimension is $d + 1$, generalization of the $\mathbb{R}^2$ case.

VC Upper bound

Vapnik & Chervonenkis

For any m , for any $\delta \in (0, 1)$, with probability (w.p.) $\geq 1 - \delta$:

$$\forall h \in \mathcal{H}, \Delta(h) \leq \sqrt{\frac{8}{m} \log \left(\frac{4G_{\mathcal{H}}(2m)}{\delta} \right)}$$

- Bounding the growth function $\rightarrow$ Sauer's Lemma
- if $d = VC(\mathcal{H})$ finite, then $G_{\mathcal{H}} \leq \sum_{k=0}^d \binom{m}{k}$ for all m which implies that $G_{\mathcal{H}}(m) \leq (em/d)^d$ (polynomial in m)

For $\mathcal{H}$ with $d = VC(\mathcal{H})$ finite, for any m , for any $\delta \in (0, 1)$, w.p. $\geq 1 - \delta$:

$$\forall h \in \mathcal{H}, \Delta(h) \leq \sqrt{\frac{8d}{m} \log \left(\frac{2em}{d} \right) + \frac{8}{m} \log \left(\frac{4}{\delta} \right)}$$

PAC learnability

VC upper bound

Note that the bound is: the same for all functions in the class (uniform over $\mathcal{H}$) and the same for all distributions (uniform over P)

VC lower bound

VC dimension characterises learnability in PAC setting: there exist distributions such that with large probability over m random examples, the gap between the risk and the best possible risk achievable over the class is at least

$$\sqrt{\frac{d}{m}}$$

Limitations of the VC framework

- The theory is certainly valid and tight - lower and upper bounds match!
- VC bounds motivate Empirical Risk Minimization (ERM), as apply to a hypothesis space, not hypothesis-dependent
- Practical algorithms often do not search a fixed hypothesis space but regularize to trade complexity with empirical error, e.g. k-NN or SVMs or DNNs
- Mismatch between theory and practice (SVM, ...)

Recap and what is coming

- Statistical Learning Theory bounds the tail of the error distribution
- giving high confidence bounds on generalization
- VC gave uniform bounds over a set of classifiers and worst-case over data-generating distributions
- VC characterizes learnability (for a fixed class)
- Fat-shattering is a generalization of the VC dimension (adapted to classes) that can be applied to more general cases with real valued functions (e.g. margin-based classifier). This notion is very related to covering numbers.
- **Coming up:** exploiting non worst-case distributions
- **Coming up:** bounds that depend on the chosen function
- **Coming up:** new proof techniques

Structural risk minimization (SRM)

First step towards non-uniform learnability

- $\mathcal{H} = \bigcup_{k \in \mathbb{N}} \mathcal{H}_k$ (countable union), each $d_k = VC(\mathcal{H}_k)$ finite.
- Use a weighting scheme: w_k weight of class $\mathcal{H}_k$, $\sum_k w_k \leq 1$
For each k , $P^m[\exists f \in \mathcal{H}_k, \Delta(f) \geq \epsilon_k] \leq w_k \delta$, then by union bound:

$$\forall k \in \mathbb{N}, \forall h \in \mathcal{H}_k, \Delta(h) \leq \epsilon_k$$

- Comments:
 - First attempt to introduce hypothesis-dependence (i.e. complexity depends on the chosen function)
 - The bound leads to a **bound-minimizing algorithm**
 $k(h) := \min\{k : h \in \mathcal{H}_k\}$, return $\arg \min_{h \in \mathcal{H}} \{\hat{\mathcal{R}}_S(h) + \epsilon_{k(h)}\}$

Detecting Benign Distributions

- SRM detects 'right' complexity for the particular problem, but must define the hierarchy a priori
- need to have more nuanced ways to detect how benign a particular distribution is
- SVM uses the margin: appears to detect 'benign' distribution in the sense that data unlikely to be near decision boundary → easier to classify

Other Framework Rademacher Complexity

- Can be interpreted as a kind of average alternative of VC dimension.

Empirical Rademacher Complexity

Let $S = \{x_1, \dots, x_n\}$ be a fixed sample of size n , the **empirical Rademacher complexity** of $\mathcal{H}$ with respect to S is defined as:

$$\mathcal{R}_m^{rad}(\mathcal{H}) = \mathbb{E}_{S \sim P^m} [\hat{R}_S^{rad}(\mathcal{H})] = \mathbb{E}_{S \sim P^m} \left[\mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(x_i) \right] \right]$$

where $\sigma = (\sigma_1, \dots, \sigma_m)^T$ with σ_i independent uniform random variables in $\{-1, 1\}$ called the Rademacher variables.

Rademacher Complexity

Rademacher Complexity

Let P the distribution according to which samples are drawn. For any $n \geq 1$, the Rademacher complexity of G is the expectation of empirical Rademacher complexity over all samples of size n drawn according to P :

$$\mathcal{R}_m^{rad}(\mathcal{H}) = \mathbb{E}_{S \sim P^m}[\hat{R}_S^{rad}(\mathcal{H})]$$

Rademacher Complexity

Start: Uniform bound (over the class) on the gap:

$$P^m[\forall h \in \mathcal{H}, \Delta(h) \leq \epsilon] = P^m[\sup_{h \in \mathcal{H}} \Delta(h) \leq \epsilon]$$

Original sample: $S = (z_1, \dots, z_m)$, $\Delta(h) = \mathcal{R}(h) - \hat{\mathcal{R}}_S(h)$

Ghost sample: $S' = (z'_1, \dots, z'_m)$, $\mathcal{R}_{S'}(h) = \mathbb{E}_{S' \sim P^m} \hat{\mathcal{R}}_{S'}(h)$

$$\begin{aligned} \mathbb{E}_{S \sim P^m}[\sup_{h \in \mathcal{H}} \Delta(h)] &\leq \mathbb{E}_{S, S' \sim P^{2m}}[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m [\ell(h, z'_i) - \ell(h, z_i)]] \\ &= \mathbb{E}_{S, S' \sim P^{2m}} \mathbb{E}_{\sigma}[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i [\ell(h, z'_i) - \ell(h, z_i)]] \\ &\leq 2 \mathbb{E}_{S \sim P^m} \mathbb{E}_{\sigma}[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i \ell(h, z_i)] \end{aligned}$$

$\sigma_i \in \{-1, 1\}$ iid symmetric Rademacher variables and by symmetrization:
supremum unchanged when we swap z_i, z'_i . Then, we sum the 2 loss terms
 $\Rightarrow$ Rademacher complexity of a class

Generalization from Rademacher Complexity

- Empirical Rademacher Complexity

$$\mathcal{R}_m^{rad} = \mathcal{R}^{rad}(\mathcal{H}, S_m) = \mathbb{E}_\sigma [\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i \ell(h, z_i)]$$

- Rademacher complexity: $\mathcal{R}^{rad}(\mathcal{H}) = \mathbb{E}_{S \sim P^m} [\mathcal{R}^{rad}(\mathcal{H}, S_m)]$

- Symmetrization: $\mathbb{E}_{S \sim P^m} [\sup_{h \in \mathcal{H}} \Delta(h)] \leq 2\mathcal{R}^{rad}(\mathcal{H})$

- McDiarmid ineq.:

$$\sup_{h \in \mathcal{H}} \Delta(h) \leq \mathbb{E}_{S \sim P^m} [\sup_{h \in \mathcal{H}} \Delta(h)] + \sqrt{\frac{1}{2m} \log(1/\delta)}$$

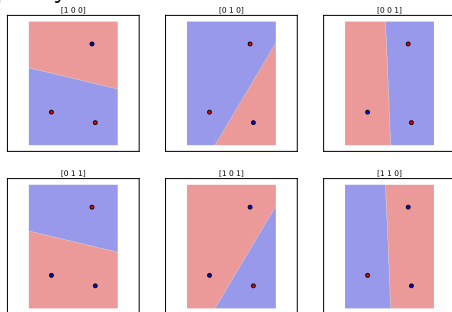
- McDiarmid ineq: $\mathcal{R}^{rad}(\mathcal{H}) \leq \mathcal{R}^{rad}(\mathcal{H}, S_m) + \sqrt{\frac{1}{2m} \log(1/\delta)}$

For any m , for any $\delta \in (0, 1)$, w.p. $\geq 1 - \delta$

$$\forall h \in \mathcal{H}, \Delta(h) \leq 2\mathcal{R}^{rad}(\mathcal{H}, S_m) + 3\sqrt{\frac{1}{2m} \log(2/\delta)}$$

Rademacher Complexity

- Ability of the considered hypotheses at fitting random noise, *i.e.* over-fitting capacity.



- Example of generalization bounds for binary classification (with a k -Lipschitz loss):

$$R(h) \leq \hat{R}(h) + 2k\mathcal{R}_m^{rad}(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2m}}; \quad R(h) \leq \hat{R}(h) + 2k\hat{\mathcal{R}}_S^{rad}(\mathcal{H}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}$$

- Empirical Rademacher complexity is sometimes hard to estimate

Example - linear classifiers

Hypothesis class $\mathcal{H}$

We consider the class of linear predictors such that the weights are bounded in L_2 -norm by a constant W_2

$$\mathcal{H} = \{h(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} \mid \|\mathbf{w}\|_2 \leq W_2\}$$

We also assume that for each instance $\mathbf{x}$ we have: $\|\mathbf{x}\| \leq X_2$.

Bounding $\hat{\mathcal{R}}_S^{rad}$

$$\begin{aligned}\hat{\mathcal{R}}_S^{rad}(h) &= \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) \right] \\&= \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{\|\mathbf{w}\|_2 \leq W_2} \sum_{i=1}^m \sigma_i \mathbf{w} \cdot \mathbf{x}_i \right] \\&= \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{\|\mathbf{w}\|_2 \leq W_2} \mathbf{w} \cdot \sum_{i=1}^m \sigma_i \mathbf{x}_i \right] \\&\leq \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{\|\mathbf{w}\|_2 \leq W_2} \left| \mathbf{w} \cdot \sum_{i=1}^m \sigma_i \mathbf{x}_i \right| \right] \\&\leq \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{\|\mathbf{w}\|_2 \leq W_2} \|\mathbf{w}\|_2 \left\| \sum_{i=1}^m \sigma_i \mathbf{x}_i \right\|_2 \right] \quad \text{Cauchy Schwarz inequality} \\&\leq \frac{W_2}{m} \mathbb{E}_\sigma \left[\left\| \sum_{i=1}^m \sigma_i \mathbf{x}_i \right\|_2 \right] \leq \frac{W_2}{m} \sqrt{\mathbb{E}_\sigma \left[\left\| \sum_{i=1}^m \sigma_i \mathbf{x}_i \right\|_2^2 \right]} \leq \frac{W_2}{m} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^m \|\sigma_i \mathbf{x}_i\|_2^2 \right]} \\&\leq \frac{W_2}{m} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^m |\sigma_i| \|\mathbf{x}_i\|_2^2 \right]} = \frac{W_2}{m} \sqrt{\left\| \sum_{i=1}^m \mathbf{x}_i \right\|_2^2} = \frac{X_2 W_2}{\sqrt{m}}\end{aligned}$$

Extension: Error bound for SVM

- Consider the class of real valued functions $\mathcal{F}(k, b)$ in a feature space defined by a kernel k with the 2-norm over the weights bounded by B : $\mathcal{R}_m^{rad}(\mathcal{H}) = \frac{B}{m} \sqrt{\sum_{i=1}^m k(x, x_i)}$
hinge loss is a lipschitz function.
- Generalization error:

$$\mathcal{R}(h) \leq \frac{1}{m\gamma} \sum_{i=1}^m \xi_i + \frac{4}{m\gamma} \sqrt{\sum_{i=1}^m k(x, x_i)} + 3\sqrt{\frac{\log(2/\delta)}{2m}}$$

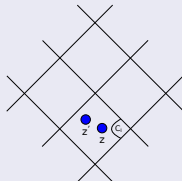
- For Gaussian kernel:

$$\mathcal{R}(h) \leq \frac{1}{m\gamma} \sum_{i=1}^m \xi_i + \frac{4}{\sqrt{m}\gamma} + 3\sqrt{\frac{\log(2/\delta)}{2m}}$$

Other frameworks

Algorithmic Robustness [Xu and Mannor, 2012]

- Consider an ϵ -cover of the input space (in binary classification: one cover for the positives, one cover for the negatives) leading to K disjoint sets.
- A learning algorithm A is $(K, \epsilon(S))$ robust if the instance space can be partitioned into K disjoint sets $\{C_i\}_{i=1}^K$ such that for all $x \in S$:
 $x, z \in C_i \Rightarrow |\text{loss}(A_S, x) - \text{loss}(A_S, z)| \leq \epsilon(S)$
→ the losses of 2 examples in the same partition must not differ too much



- Example of generalization bound (with loss function bounded by M):

$$R(h) \leq \hat{R}(h) + \epsilon(S) + M \sqrt{\frac{2K \ln(2) + 2 \ln(1/\delta)}{m}}$$

- Bounds often loose because K is large ($K \sim$ covering number exponential in the dimension), but easy to derive. Advantage: depends on the algorithm!

The PAC-Bayes framework: majority votes [Germain et al., 2015]

- Before data, fix a distribution on Q_0 on $\mathcal{H}$: **prior**
- Based on data, learn a distribution Q on $\mathcal{H}$: **posterior**
- Predictions:
 - draw $h \sim Q$ and predict with chosen h
 - each prediction with a fresh random draw
- Empirical Risk: $\hat{\mathcal{R}}_S(Q) = \int_{\mathcal{H}} \hat{\mathcal{R}}_S dQ(h)$
- Generalization risk: $\mathcal{R}(Q) = \int_{\mathcal{H}} \mathcal{R}(h) dQ(h)$
- **Classic PAC-Bayes bound**: Fix Q_0 , for any sample size m , for any $\delta \in (0, 1)$, w.p. $\geq 1 - \delta$

$$|\hat{\mathcal{R}}_S(Q) - \mathcal{R}(Q)| \leq \sqrt{\frac{1}{2m} \left[KL(Q \| Q_0) + \ln \left(\frac{2\sqrt{m}}{\delta} \right) \right]}$$

with $KL(Q, P) = \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} dx$; in discrete case (e.g. fix number of classifiers): $KL(Q, P) = \sum_i p(i) \log \frac{p(i)}{q(i)}$

PAC-Bayes bound vs Bayesian learning [Germain et al.,2015]

- Prior
 - PAC-Bayes bounds: bounds hold even if prior incorrect
 - Bayesian: inference must assume prior is correct
- Posterior
 - PAC-Bayes bounds: bound holds for all posteriors
 - Bayesian: posterior computed by Bayesian inference
- Data distribution
 - PAC-Bayes bounds: can be used to define prior, hence no need to be known explicitly
 - Bayesian: input effectively excluded from the analysis: randomness in the noise model generating the output

Typical form of PAC-Bayes bounds (D a divergence):

$$\forall Q, D(\hat{\mathcal{R}}_S(Q) || \mathcal{R}(Q)) \leq \frac{KL(\hat{\mathcal{R}}_S(Q_0) || \mathcal{R}(Q_0)) + \log(\frac{I(m)}{\delta})}{m}$$

PAC-Bayes bound for SVMs

- Let α_m be the set of weights learned from a sample of size m , let $\hat{\alpha}_m = \alpha_m / \|\alpha_m\|$
- When using Gaussian randomization $Q_0 = \mathcal{N}(0, I)$, $Q_\mu = \mathcal{N}(\mu\alpha_m, I)$ then: $KL(Q_\mu \| Q_0) = \frac{1}{2}\mu^2$.
Then have for any m , any $\delta \in (0, 1)$, w.p. $\geq 1 - \delta$:

$$KL(\hat{\mathcal{R}}_S(Q) \| \mathcal{R}(Q)) \leq \frac{\frac{1}{2}\mu^2 + \log \frac{m+1}{\delta}}{m}$$

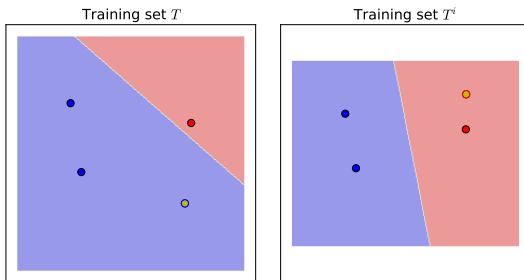
- SVM generalization error $\leq 2\min_{\mu} \mathcal{R}(Q_\mu)$

$\Rightarrow$ PAC-bayesian bounds can be very tight, but requires to express a classifier under a majority-vote based scheme, the prior must be set (otherwise generally taken as uniform over the class), it also induces some technical aspect.

Algorithmic stability

Principle

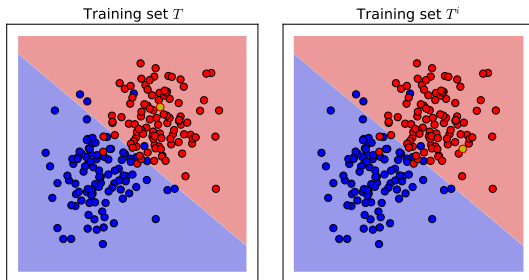
Intuitively, an algorithm A is said stable if it is robust to small changes in the training sample, i.e., the variation in its output h is small w.r.t the sample size.



Algorithmic stability

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Algorithmic stability

Principle

Intuitively, an algorithm A is said stable if it is robust to small changes in the training sample, i.e., the variation in its output h is small w.r.t the sample size.

- Uniform stability has the nice property to provide generalization guarantees with respect to the algorithm (formulated as an optimization problem) used to learn the model
- In opposition to other framework (VC-dimension, Rademacher complexity) that gives generalization bound related to the capacity of the hypothesis class $\mathcal{H}$

We will focus on a particular case: the **uniform stability**

Uniform stability - formal definition

Given a training set S of size m , we build $\forall i = 1, \dots, m$:

- $S^{\setminus i} = \{z_1, \dots, z_{i-1}, z_{i+1}, \dots, z_m\}$ by removing the i -th element of S
- $S^i = \{z_1, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_m\}$ by replacing the i -th element of S by z'_i drawn i.i.d. from P

Definition

An algorithm L has uniform stability $\frac{\beta}{m}$ w.r.t a loss function ℓ if the following holds:

$$\forall S, \forall i \in \{1, \dots, m\}, \sup_z |\ell(h_S, z) - \ell(h_{S^{\setminus i}}, z)| \leq \frac{\beta}{m}$$

where β is a positive constant, h_S and $h_{S^{\setminus i}}$ are the hypothesis learned by L from S and $S^{\setminus i}$ respectively

$$\begin{aligned} \text{Notations: } \hat{\mathcal{R}}^i &= \frac{1}{m} \sum_{j=1}^m \ell(h_S, z_j), \quad \hat{\mathcal{R}}^{\setminus i} = \frac{1}{m} \sum_{j=1, j \neq i}^m \ell(h_{S^{\setminus i}}, z_j), \\ \hat{\mathcal{R}}^i &= \frac{1}{m} \sum_{z_j \in S^i} \ell(h_{S^i}, z_j) = \frac{1}{m} \sum_{j=1, j \neq i}^m \ell(h_{S^i}, z_j) + \ell(h_{S^i}, z'_i) \end{aligned}$$

Generalization bound using uniform stability

Theorem: Generalization Bound from uniform stability

Let S be a training sample of size m and $\delta > 0$. For any algorithm L with uniform stability $\frac{\beta}{m}$ with respect to a loss function ℓ bounded by M , with probability $1 - \delta$, we have:

$$\mathcal{R}_{h_S} \leq \hat{\mathcal{R}}_{h_S} + \frac{2\beta}{m} + (4\beta + M) \sqrt{\frac{\ln 1/\delta}{2m}}.$$

- From [Kearns and Ron, 1999], an algorithm applied on an hypothesis space of finite VC-dimension is stable in the sense that its stability is bounded by its VC-dimension
- Using stability as a complexity measure does not give worse bounds than VC-dim bounds

Proof of generalization bound - based on McDiarmid's Thm

Theorem - McDiarmid's (1989)

Let S and S^i defined as above, let $F : Z^m \rightarrow \mathbb{R}$ be a function for which there exists constants c_i ($i = 1, \dots, m$) such that:

$$\sup_{S \in Z^m, z'_i} |F(S) - F(S^i)| \leq c_i,$$

then

$$P_S[F(S) - \mathbb{E}_S[F(S)] \geq \epsilon] \leq \exp(-2\epsilon^2 / \sum_{i=1}^m c_i^2)$$

This a generalization of the Hoeffding bound.

We use $F(S) = \mathcal{R}(h) - \hat{\mathcal{R}}(h)$ - denoted as $F(S) = \mathcal{R} - \hat{\mathcal{R}}$ for sake of simplicity - and $F(S^i) = \mathcal{R}^i - \hat{\mathcal{R}}^i$

Proof of Generalization Bound

By definition, the algorithm is β stable, therefore we have:

$$|\mathcal{R} - \mathcal{R}^{\setminus i}| = |\mathbb{E}_z[\ell(h_S, z)] - \mathbb{E}_z[\ell(h_{S \setminus i}, z)]| \leq \mathbb{E}_z|\ell(h_S, z) - \ell(h_{S \setminus i}, z)| \leq \frac{\beta}{m}$$

We deduce from triangle inequality that:

$$|\mathcal{R} - \mathcal{R}^i| \leq |\mathcal{R} - \mathcal{R}^{\setminus i}| + |\mathcal{R}^{\setminus i} - \mathcal{R}^i| \leq 2\frac{\beta}{m}$$

Moreover,

$$\begin{aligned} |\hat{\mathcal{R}} - \hat{\mathcal{R}}^{\setminus i}| &= \left| \frac{1}{m} \sum_{j \neq i} (\ell(h_S, z_j) - \ell(h_{S \setminus i}, z_j)) + \frac{1}{m} \ell(h_S, z_i) \right| \\ &\leq \frac{1}{m} \sum_{j \neq i} |\ell(h_S, z_j) - \ell(h_{S \setminus i}, z_j)| + \frac{1}{m} |\ell(h_S, z_i)| \\ &\leq \frac{\beta}{m} + \frac{M}{m} \end{aligned}$$

We use stability hypothesis for 1st element and the fact that the loss is bounded by M for the second.

Proof of Generalization Bound

From $|\hat{\mathcal{R}} - \hat{\mathcal{R}}^{\setminus i}| \leq \frac{\beta+M}{m}$, we deduce with triangle inequality:

$$|\hat{\mathcal{R}} - \hat{\mathcal{R}}^i| \leq |\hat{\mathcal{R}} - \hat{\mathcal{R}}^{\setminus i}| + |\hat{\mathcal{R}}^{\setminus i} - \hat{\mathcal{R}}^i| \leq 2\frac{\beta+M}{m}$$

However, a close look reveals that 2nd factor 2 is not needed:

$$\begin{aligned} |\hat{\mathcal{R}} - \hat{\mathcal{R}}^i| &\leq \frac{1}{m} \sum_{j \neq i} |\ell(h_S, z_j) - \ell(h_{S^i}, z_j)| + \frac{1}{m} |\ell(h_S, z_i) - \ell(h_{S^i}, z'_i)| \\ &\leq \frac{1}{m} \sum_{j \neq i} |\ell(h_S, z_j) - \ell(h_{S \setminus i}, z_j)| + \frac{1}{m} \sum_{j \neq i} |\ell(h_{S \setminus i}, z_j) - \ell(h_{S^i}, z_j)| \\ &\quad + \frac{1}{m} |\ell(h_S, z_i) - \ell(h_S, z'_i)| \\ &\leq 2\frac{\beta}{m} + \frac{M}{m} \end{aligned}$$

Considering the random variable $F(S) = \mathcal{R} - \hat{\mathcal{R}}$, we deduce :

$$\begin{aligned} |F(S) - F(S^i)| &= |(\mathcal{R} - \hat{\mathcal{R}}) - (\mathcal{R}^i - \hat{\mathcal{R}}^i)| \\ &\leq |(\mathcal{R} - \mathcal{R}^i)| + |(\hat{\mathcal{R}} - \hat{\mathcal{R}}^i)| \\ &\leq 4 \frac{\beta}{m} + \frac{M}{m} \\ &= c_i \text{ (of McDiarmid's Thm)} \end{aligned}$$

This informs us that the variable $\mathcal{R} - \hat{\mathcal{R}}$ satisfies McDiarmid's conditions. It remains to bound the expectation of this random variable to apply the following bound:

$$P_S[F(S) - \mathbb{E}_S[F(S)] \geq \epsilon] \leq \exp(-2\epsilon^2 / \sum_{i=1}^m c_i^2)$$

Proof of Generalization Bound

$$\begin{aligned}\mathbb{E}_S[\mathcal{R} - \hat{\mathcal{R}}] &= \mathbb{E}_{S, z'_i}[\ell(h_S, z'_i)] - \mathbb{E}_S[\hat{\mathcal{R}}] \\&= \mathbb{E}_{S, z'_i}[\ell(h_S, z'_i)] - \frac{1}{m} \sum_{j=1}^m \mathbb{E}_S[\ell(h_S, z_j)] \\&= \mathbb{E}_{S, z'_i}[\ell(h_S, z'_i)] - \frac{1}{m} \sum_{j=1}^m \mathbb{E}_{S, z'_i}[\ell(h_{S^i}, z'_i)] \text{ (it holds } \forall S \in Z^m) \\&= \mathbb{E}_{S, z'_i}[\ell(h_S, z'_i) - \ell(h_{S^i}, z'_i)] \\&\leq \mathbb{E}_{S, z'_i}[|\ell(h_S, z'_i) - \ell(h_{S^i}, z'_i)|] \\&\leq \mathbb{E}_{S, z'_i}[|\ell(h_{S^i}, z'_i) - \ell(h_{S \setminus i}, z'_i)|] + \mathbb{E}_{S, z'_i}[|\ell(h_{S \setminus i}, z'_i) - \ell(h_S, z'_i)|] \\&\text{(now since } \sup_{S, z} |\ell(h_S, z) - \ell(h_{S \setminus i}, z)| \leq \frac{\beta}{m} \text{ by uniform stability)} \\&\text{then } \Rightarrow \mathbb{E}_{S, z}[|\ell(h_S, z) - \ell(h_{S \setminus i}, z)|] \leq \frac{\beta}{m} \\&\leq 2 \frac{\beta}{m}\end{aligned}$$

Proof of Generalization Bound

McDiarmid's Thm tells us that

$$P_S[F(S) - \mathbb{E}_S[F(S)] \geq \epsilon] \leq \exp(-2\epsilon^2 / \sum_{i=1}^m c_i^2)$$

If we set $F(S) = \mathcal{R} - \hat{\mathcal{R}}$, we get

$$P_S \left[\mathcal{R} - \hat{\mathcal{R}} - \mathbb{E}_S[F(S)] > \epsilon \right] \leq \exp \left(\frac{-2\epsilon^2}{m(4\frac{\beta}{m} + \frac{M}{m})^2} \right) = \exp \left(\frac{-2m\epsilon^2}{(4\beta + M)^2} \right)$$

Thus, setting the right hand side to δ , using the fact that $\mathbb{E}_S[F(S)] \leq \frac{2\beta}{m}$, we get with probability at least $1 - \delta$

$$\mathcal{R} \leq \hat{\mathcal{R}} + 2\frac{\beta}{m} + (4\beta + M)\sqrt{\frac{\ln(1/\delta)}{2m}}$$

which gives the bound.

Stability Constant and Generalization Bound

Bound and Stability constant

$$\mathcal{R} \leq \hat{\mathcal{R}} + \frac{2\beta}{m} + (4\beta + M)\sqrt{\frac{\ln 1/\delta}{2m}}.$$

This upper bound tells us that if we can find the stability constant β of an algorithm A , then the learned hypothesis has a generalization guarantee.

How to find this constant?

How to find the β constant

Stability with regularizers of quadratic form

For problem of this type: $\min_{h \in \mathcal{H}} \frac{1}{m} \sum_i^m \ell(h, z_i) + \lambda \|h\|^2$

From [Bousquet and Elisseeff 2002], the stability bound can be defined as follows:

$$\beta \leq \frac{\sigma^2}{2\lambda}$$

σ is an admissibility constant of the cost function

Uniform stability works well with learning problems with regularizers of “quadratic” form, but not for sparsity inducing regularizers (L_1 norm for example) other frameworks must be used in that case (Rademacher, Robustness, ...)

Extra Material: exercise sheet

Exercise sheet on uniform stability

- Let $L = \{z_j = (x_j, y_j)\}_{j=1}^m$ with $x_j \in \mathcal{X}$ and $y_j \in [0, B] = \mathcal{Y}$, with $\|x_j\| \leq K$
- We consider the following ridge regression problem:
$$\min_{\theta} \sum_{j=1}^m \ell(\theta, z_j) + \lambda \|\theta\|_2^2$$
- With the loss $\ell(\theta, z_j) = (\theta^T x_j - y_j)^2 = c(\theta^T x, y)$
- Then we can get $\sigma = 2 * (B + \frac{BK}{\sqrt{\lambda}})$ and thus

$$\beta = \frac{\sigma^2}{2\lambda} = \frac{2\left(B + \frac{BK}{\sqrt{\lambda}}\right)^2}{\lambda}$$

- Denoting $\mathcal{R}_{\theta}$ as the true risk, we can get the final final bound:

$$\mathcal{R}_{\theta} \leq \hat{\mathcal{R}}_{\theta} + \frac{2\sigma^2 K^2}{\lambda m} + \left(\frac{4\sigma^2 K^2}{\lambda} + M \right) \sqrt{\frac{\ln 1/\delta}{2m}}$$

Additional material Solutions of the sheet

2- σ -admissibility

Pretty easy, if we assume each $y_1, y_2, y \in [0, B]$

$$|\ell(y_1, y) - \ell(y_2, y)| = |(y_1 - y)^2 - (y_2 - y)^2| \quad (6)$$

$$= |(y_1 - y_2)(y_1 + y_2 - 2y)| \quad (7)$$

$$\leq 2B|y_1 - y_2| \quad (8)$$

Upper bound for $\|\Delta\theta\|$

4.1 We can apply Lemma one with $t = 1/2$, we get then

$$\|\theta\|_2^2 - \|\frac{1}{2}(\theta + \theta^{\setminus i})\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\frac{1}{2}(\theta^{\setminus i} + \theta)\|_2^2 \leq \frac{1\sigma}{2\lambda m} |(\Delta\theta)^T x_i|.$$

4.2 We have

$$\|\theta\|_2^2 - \|\frac{1}{2}(\theta + \theta^{\setminus i})\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\frac{1}{2}(\theta^{\setminus i} + \theta)\|_2^2 = \frac{1}{2}\|\Delta\theta\|^2$$

and thus

$$\frac{1}{2}\|\Delta\theta\|_2^2 \leq \frac{\sigma}{2\lambda m} |(\Delta\theta)^T x_i|$$

4.3 Now from Hölder inequality we have

$$|(\Delta\theta)^T x_i| \leq \|\Delta\theta\|_2 \|x_i\|_2$$

4.4 We need to take the assumption that $\|x_i\|_2 \leq K$, then by combining the previous results we have

$$\|\Delta\theta\|_2 \leq \frac{\sigma K}{\lambda m}$$

4.5 Bound on $\|\theta\|_2^2$

If θ is a minimizer of $R_r(\cdot)$ then, for the null hypothesis $\mathbf{0}$ the associated risk R_r must be higher than the one associated to θ : $R_r(\theta) \leq R_r(\mathbf{0})$, then we must have

$$R_r(\theta) \leq R_r(\mathbf{0}) = \frac{1}{m} \sum_{j=1}^m y_i^2 \leq B^2$$

Thus:

$$\lambda \|\theta\|_2^2 \leq B^2 - \frac{1}{m} \sum_{j=1}^m (\theta^T x_i - y_i)^2 \leq B^2$$

Since $\frac{1}{m} \sum_{j=1}^m (\theta^T x_i - y_i)^2 \geq 0$, we have

$$\|\theta\|_2^2 \leq \frac{B^2}{\lambda}$$

4.6 Bound on the loss

If θ has been learned from a ridge regression problem: $\|\theta\| \leq \frac{B}{\sqrt{\lambda}}$

For the ℓ , using Hölder inequality for 2nd inequality, we have for any z :

$$\ell(\theta, z) = (\theta^T x - y)^2 \leq (|\theta^T x| + B)^2 \leq \left(\frac{BK}{\sqrt{\lambda}} + B\right)^2 = B^2 \frac{(K + \sqrt{\lambda})^2}{\lambda} = M$$

Note: the particular case of Hölder inequality where $p = q = 2$ is called the Cauchy-Schwartz inequality.

5.1 and 5.2 - Stability constant and Generalization bound

5.1 We have for any z

$$|\ell(\theta, z) - \ell(\theta^{\setminus i}, z)| \leq \sigma |(\Delta\theta)^T x| \quad (9)$$

$$\leq \sigma \frac{\sigma K}{\lambda m} K \quad (10)$$

$$= \frac{\sigma^2 K^2}{\lambda m} \quad (11)$$

(9) comes from the σ -admissibility, (10) uses Hölder inequality and the bound on $\|\Delta\theta\|$ and $\|x\| \leq K$ twice.

5.2 From the stability theorem, we have

$$\mathcal{R}_\theta \leq \hat{\mathcal{R}}_\theta + \frac{2\sigma^2 K^2}{\lambda m} + \left(\frac{4\sigma^2 K^2}{\lambda} + M \right) \sqrt{\frac{\ln 1/\delta}{2m}}.$$

Sometimes the examples are normalized to ensure $K = 1$.

5.3 More General result - Improving σ

Assuming $y_1, y_2 \in [0, B]$ is strong and not true in practice, we will refine this later to be true for any learned model θ . Actually, using the bounds on $\|\theta\|$ of page 34, the assumption that $\|x\|_2 \leq K$ (cf later) and Hölder inequality, we can get that $|\theta^T x| \leq \|\theta\|_2 \|x\|_2 \leq \frac{BK}{\sqrt{\lambda}}$. Then the $|y_i - y| \leq |y_i| + |y|$ ($i = 1, 2$) can then be bounded by $\frac{BK}{\sqrt{\lambda}} + B$ since the y_i have the form $\theta^T x_i$, and thus $\sigma = 2 * (B + \frac{BK}{\sqrt{\lambda}})$ is a better choice. Since the result proven previously is very general, it is easy to update the bound with the new σ .

5.4 More general result - using norms

Replace $\|\cdot\|_2$ replace by $N(\cdot)$ any proper closed convex and differentiable regularizer, ℓ any bounded convex loss function. The result of Lemma 1 can be generalized and thus applied directly for a broad family of losses (and regularizers).

Appendix - proof of Lemma 1

Since c is convex, $\mathcal{R}^{\setminus i}$ is convex and thus we have for any $t \in [0, 1]$

$$\hat{\mathcal{R}}^{\setminus i}(\theta + t\Delta\theta) - \hat{\mathcal{R}}^{\setminus i}(\theta) \leq t(\hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i}) - \hat{\mathcal{R}}^{\setminus i}(\theta))$$

Switching the role of θ and $\theta^{\setminus i}$

$$\hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i} - t\Delta\theta) - \hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i}) \leq t(\hat{\mathcal{R}}^{\setminus i}(\theta) - \hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i}))$$

Summing the two preceding inequalities yields

$$\hat{\mathcal{R}}^{\setminus i}(\theta + t\Delta\theta) - \hat{\mathcal{R}}^{\setminus i}(\theta) + \hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i} - t\Delta\theta) - \hat{\mathcal{R}}^{\setminus i}(\theta^{\setminus i}) \leq 0 \quad (12)$$

By assumption, we have

$$R_r(\theta) - R_r(\theta + t\Delta\theta) \leq 0, \quad R_r^{\setminus i}(\theta^{\setminus i}) - R_r^{\setminus i}(\theta^{\setminus i} - t\Delta\theta) \leq 0$$

Summing the two previous inequalities and using (12), we have:

$$c(\theta^T x_i, y_i) - c((\theta + t\Delta\theta)^T x_i, y_i) + m\lambda(\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2) \leq 0$$

Appendix - proof of Lemma 1 (continuing)

The previous line can be rewritten as:

$$m\lambda(\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2) \leq c((\theta + t\Delta\theta)^T x_i, y_i) - c(\theta^T x_i, y_i)$$

and then

$$m\lambda(\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2) \leq |c((\theta + t\Delta\theta)^T x_i, y_i) - c(\theta^T x_i, y_i)|$$

Thus

$$\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2 \leq \frac{1}{m\lambda} |c((\theta + t\Delta\theta)^T x_i, y_i) - c(\theta^T x_i, y_i)|$$

By the σ -admissibility condition, we get

$$\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2 \leq \frac{t\sigma}{\lambda m} |(\Delta\theta)^T x_i|.$$